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# Modulational instability of plasma waves from a staggered $ZZ$ interaction in the spin-chain plasma-wave model

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## Abstract

A recent experiment demonstrated that the linear dynamics of electromagnetic waves in a plasma can be simulated on a superconducting quantum processor by mapping the wave equation onto a nearest-neighbor spin chain and evolving singly-excited states. That construction is restricted to the single-excitation sector, where the model is free (quadratic) and therefore classically simulable in polynomial time; the physically rich regime of nonlinear laser–plasma interaction lies outside it. Here we take the next step proposed in that work: we add a nearest-neighbor, bond-alternating (staggered)  $ZZ$  interaction to the spin-chain Hamiltonian and show that it supplies the cubic self/cross-phase coupling responsible for the modulational instability (MI) of a plasma wave. We derive the mapping analytically: in the dilute-magnon (Holstein–Primakoff) and classical-correspondence limit the staggered  $ZZ$  term reduces to the focusing cubic nonlinearity of a nonlinear Schrödinger equation (NLSE), and its staggered sign is precisely what makes the four-wave-mixing vertex phase-match across the two branches of the Dirac-like dispersion. We then emulate the full  $2^{12}$ -dimensional quantum dynamics with a Trotterized statevector simulator and compare the resonant four-wave-mixing sideband-conversion coefficient  $C$  against an independently integrated classical spin-coherent (Landau–Lifshitz) reference. Across a pre-registered five-point sweep of the coupling  $\epsilon \in [0.1, 0.5]$ , the emulation and reference agree to 10–12%, inside the pre-stated 15% tolerance, with the unextended model showing zero sideband conversion. The agreement is limited by the hard-core (quantum) correction to the classical reference, which we isolate and discuss.

## 1 Introduction

Simulating how electromagnetic waves propagate and scatter in plasmas is a computational bottleneck in problems ranging from inertial-confinement fusion to ionospheric physics. Sundar *et al.* [1] recently showed that the *linear* plasma-wave equation can be encoded in a one-dimensional spin chain whose single-excitation sector reproduces the plasma dispersion  $\omega_k = \pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$ , and they demonstrated propagation of a wave packet on a superconducting device. The power of the mapping is that the spin model is *not* tied to the linear regime: the authors explicitly note that adding  $ZZ$  interactions takes the model out of the exactly solvable single-particle class and opens the door to “nonlinear effects in plasmas, such as electromagnetically induced transparency, laser-plasma scattering, and modulational instabilities.”

This paper pursues that invitation for one specific and well-characterized nonlinear phenomenon: the *modulational instability* of a plasma wave, the process by which a monochromatic carrier wave becomes unstable to the growth of sidebands and breaks up into envelopes. In the envelope limit MI is

governed by the focusing cubic nonlinear Schrödinger equation [2, 3, 4, 5], and its earliest signature is resonant four-wave mixing (FWM): two pump photons (magnons) at the carrier frequency scatter into a signal and an idler sideband. The FWM sideband-conversion rate and the MI gain are the same cubic nonlinearity viewed in the spectral and time domains, so a clean, quantitative observable of the onset of MI is the early-time sideband-conversion coefficient.

Our contribution is threefold. (i) We identify a specific, physically motivated  $ZZ$  coupling structure—nearest-neighbor and *staggered* (bond-alternating sign)—and show analytically that it realizes the cubic NLSE nonlinearity on this lattice, whereas a uniform  $ZZ$  coupling does not (Sec. 2 and 3). (ii) We emulate the full quantum dynamics on the  $2^{12}$  Hilbert space with a Trotterized statevector simulator and extract the FWM conversion coefficient  $C$  as a function of the coupling  $\epsilon$  (Sec. 4). (iii) We compare  $C$  point-by-point against an independently computed classical spin-coherent reference—the lattice precursor of the NLSE—and demonstrate agreement within a pre-stated tolerance across the whole sweep, with a controlled  $H_0$  contrast showing the effect originates from the  $ZZ$  term (Sec. 5). We close with a mechanism, failure, and computational-cost analysis (Sec. 6).

## 2 The model

**Base Hamiltonian.** We work with the spin-chain plasma-wave model of Ref. [1],

$$H_0 = \frac{J}{4} \sum_{j=0}^{N-2} (X_j Y_{j+1} - Y_j X_{j+1}) + \frac{1}{2} \sum_{j=0}^{N-1} \Delta_j (-1)^{j+1} Z_j, \quad (1)$$

where  $X_j, Y_j, Z_j$  are Pauli operators on qubit  $j$ ,  $J = 1$  sets the energy unit, and  $\hbar = 1$ . Equation (1) is the paper's Eq. (3) up to a fixed  $\pi$ -rotation of every qubit about the  $x$  axis (a global change of basis,  $Z_j \rightarrow -Z_j$ ); it reproduces the paper's single-excitation equation of motion (their Eq. (4)), the dispersion, and the propagation dynamics exactly. The ground state  $|0 \cdots 0\rangle$  is the magnon vacuum, and a single flipped spin  $|\cdots 1_j \cdots\rangle$  is one magnon. Because the  $XY$  term is number conserving, the single-magnon sector is closed and maps onto a free-boson quadratic form whose Bloch Hamiltonian on the two-site unit cell is  $H(k) = \Delta \sigma_z + J \sin(ka) \sigma_y$ , giving  $\omega_k = \pm \sqrt{\Delta^2 + J^2 \sin^2 ka}$  (derived symbolically in Appendix A).

**Staggered  $ZZ$  extension.** We add

$$H_{ZZ} = \epsilon \sum_{j=0}^{N-2} (-1)^j Z_j Z_{j+1}, \quad K_{j,j+1} = K_{j+1,j} = (-1)^j, \quad (2)$$

and zero otherwise. The coupling is *nearest-neighbor only* (local), with a bond-alternating sign. The full Hamiltonian is  $H = H_0 + H_{ZZ}$ . Two properties are immediate. First,  $\epsilon$  is an overall multiplicative factor, so  $H \rightarrow H_0$  exactly as  $\epsilon \rightarrow 0$ : the linear model is recovered by construction. Second, under the Holstein–Primakoff identification  $Z_j = 1 - 2n_j$  (with  $n_j$  the magnon number),  $H_{ZZ}$  becomes a bond-alternating density–density coupling  $\epsilon \sum_j (-1)^j (1 - 2n_j)(1 - 2n_{j+1})$  whose density-dependent part is a modulation-wavenumber- $\pi$  interaction. As we show below, that  $\pi$  modulation is exactly what phase-matches a four-wave-mixing coupling between a carrier mode and its sidebands across the two branches of the dispersion.

**Physical motivation.** A uniform (non-staggered) nearest-neighbor  $ZZ$  term produces only a Hartree mean-field frequency shift and no resonant sideband coupling in this band structure: its quadratic part is a spatially uniform renormalization of  $\Delta$ , which shifts all modes rigidly and couples nothing. The staggered sign is the essential ingredient. It is the lattice analog of the ponderomotive, density-dependent self/cross-phase coupling that drives MI of a Langmuir wave: the nonlinearity scatters a wave at wavenumber  $k$  into  $k + \pi$ , i.e. across the electric-field and current (upper and lower branch) components of the Dirac-like dispersion, which is precisely the phase-matched channel for parametric four-wave mixing. We considered and rejected uniform  $ZZ$ , next-nearest-neighbor, and staggered next-nearest-neighbor couplings for exactly this reason (Sec. 6).

### 3 Mapping to the nonlinear Schrödinger equation

We establish the mapping in three steps; the algebra is carried out symbolically and checked numerically in `mapping_derivation.py` (Appendix A).

**1. Free part.** The quadratic part of  $H$  is a number-conserving free-boson form. On the two-site unit cell its Bloch Hamiltonian is  $H(k) = \Delta\sigma_z + J\sin(ka)\sigma_y$ , with eigenfrequencies  $\omega_k = \pm\sqrt{\Delta^2 + J^2\sin^2 ka}$ . This is the linear plasma-wave dispersion, with the two branches playing the role of the electric-field and current components.

**2. The staggered Hartree term vanishes.** Writing  $Z_j = 1 - 2n_j$ ,

$$Z_j Z_{j+1} = 1 - 2(n_j + n_{j+1}) + 4n_j n_{j+1}. \quad (3)$$

The *quadratic* (Hartree) part of  $H_{ZZ}$  is  $-2\epsilon \sum_j (-1)^j (n_j + n_{j+1})$ . In the bulk, the coefficient of each  $n_j$  collects a  $(-1)^j$  from bond  $j$  and a  $(-1)^{j-1}$  from bond  $j-1$ :

$$(-1)^j + (-1)^{j-1} = 0. \quad (4)$$

The staggered quadratic shift therefore cancels identically in the bulk, so the staggered  $ZZ$  term contributes *no* bare frequency renormalization—in sharp contrast to a uniform  $ZZ$ , whose Hartree term is  $-4\epsilon \sum_j n_j$ . This cancellation is what keeps the linear dispersion intact while still supplying a nonlinearity, and it is unique to the staggered sign.

**3. The surviving quartic vertex transfers momentum  $\pi$ .** The surviving density-dependent term is the quartic vertex

$$H_{ZZ} \supset 4\epsilon \sum_j (-1)^j n_j n_{j+1}. \quad (5)$$

Fourier transforming  $n_j = \frac{1}{M} \sum_{k,k'} e^{i(k'-k)j} a_{k'}^\dagger a_k$ , the factor  $(-1)^j = e^{i\pi j}$  makes the vertex conserve lattice momentum only modulo  $\pi$ : it scatters a pair with total momentum  $K$  into  $K + \pi$ . We verify on a small ring that  $\sum_j (-1)^j e^{i\delta k j} = M$  if and only if  $\delta k = \pi$ . A carrier magnon at the band edge  $k_0 = 0$  (lower branch,  $\omega = -\Delta$  at  $k = 0$ ) is thereby coupled to its  $\pi$ -shifted partner and, through the band curvature, to the neighboring same-branch sideband modes—the phase-matched FWM channel. Equation (5) is a local density-squared interaction, i.e. a *cubic* self/cross-phase term for the envelope field. In the continuum, low-density, classical-correspondence limit the magnon envelope  $A(x, t)$  obeys the focusing cubic NLSE

$$i \partial_t A = \frac{\omega''(k_0)}{2} \partial_x^2 A + q |A|^2 A, \quad q \propto \epsilon, \quad (6)$$

with  $\omega''(k_0) < 0$  and  $q < 0$  (focusing) for the lower branch, so the standard MI criterion  $\omega'' q > 0$  is met. This is the Taniuti–Washimi/Hasegawa result that a Langmuir wave is modulationally unstable [2, 3].

**Headline observable.** The earliest, cleanly measurable signature of the cubic coupling is resonant FWM sideband conversion. With an unseeded carrier (band-edge mode) prepared at weak occupation and the sideband initially empty, the staggered- $ZZ$  vertex drives the sideband; its population grows quadratically at short time,

$$P_{\text{side}}(t) = C t^2 + \dots, \quad C = g_{\text{FWM}}^2, \quad (7)$$

with the FWM rate  $g_{\text{FWM}} \propto \epsilon n_{\text{pump}}$ , so that  $C \propto \epsilon^2$ . We take  $C$  as the headline observable and extract it as the mean of  $P_{\text{side}}(t)/t^2$  over the early quadratic window  $t \in [0.5, 2.0]$ . The classical prediction  $C_{\text{ref}}$  is obtained by integrating the spin-coherent (Landau–Lifshitz) equations of motion of  $H = H_0 + H_{ZZ}$ —the classical (factorized) equations that the quantum model reduces to in the dilute-magnon correspondence limit and the lattice precursor of the NLSE (6)—from the *same* product-state initial condition and extracting  $C$  identically. The reference uses no data from the quantum emulation.

## 4 Method

**Quantum emulation.** We simulate the full  $2^N$ -dimensional dynamics with  $N = 12$  (4096 dimensions) using a Trotterized statevector propagator. Each step applies  $U = U_{ZZ} U_Z U_{XY}^{\text{even}} U_{XY}^{\text{odd}}$  where  $U_{XY}$  gates are the exact bond exponentials of the  $XY$  term (a  $\sqrt{\text{iSWAP}}$ -like two-qubit gate),  $U_Z$  the exact single-qubit staggered-field rotations, and  $U_{ZZ}$  the exact staggered- $ZZ$  rotations. We use a Trotter step  $\delta = 0.05$  for the nonlinear sweep (the  $N = 9$  linear baseline uses the paper’s  $\delta = 0.8$  for direct comparison). The smaller step is required because  $C$  is read off the early window  $t \in [0.5, 2.0]$ , which a coarse step under-resolves, and because it suppresses the Trotter transient that otherwise contaminates the weak sideband signal and the  $\epsilon = 0$  contrast; the step convergence is quantified in Sec. 5 and disclosed here. No single-excitation truncation is used anywhere: at every step we record the statevector norm, the total magnon number, and the population transferred outside the single-excitation subspace.

**Initial condition and modes.** The carrier is the band-edge ( $k_0 = 0$ ) lower-branch mode ( $\omega = -0.514$  for  $\Delta = 0.5$ ), prepared as a spin-coherent product state with per-site pump amplitude  $A_{\text{pump}} = 0.2$  (per-site density  $\sim 0.04$ ), chosen so the spin-coherent classical reference is quantitatively accurate; the residual offset is the hard-core correction reported below. The sideband is the nearest same-branch mode ( $\omega = -0.613$ ), initially unpopulated.

**Sweep and tolerance.** We pre-registered the five-point sweep  $\epsilon \in \{0.10, 0.20, 0.30, 0.40, 0.50\}$ , the headline observable  $C$ , the fit window, and the agreement criterion—relative error  $\leq 15\%$  on  $C$  at each point—in `method_plan.json.md` before any validation run. At every sweep point we also evolve the unextended  $H_0$  ( $\epsilon = 0$ ) under identical conditions as a contrast. All five declared points are reported.

**Classical reference.** The reference integrates the Landau–Lifshitz equations  $\dot{\mathbf{s}}_j = -2\mathbf{h}_j \times \mathbf{s}_j$  for the spin-coherent expectations  $\mathbf{s}_j = \langle \boldsymbol{\sigma}_j \rangle$ , with local field  $h_j^z = \Delta(-1)^j + \epsilon(-1)^j s_{j+1}^z + \epsilon(-1)^{j-1} s_{j-1}^z$  and the  $XY$  hopping contributing to  $h_j^x, h_j^y$ . These are integrated with `scipy.integrate.solve_ivp` at tight tolerance from the same initial condition, and  $C$  is extracted identically.

## 5 Results

**Linear limit and convergence.** At  $\epsilon = 0$  the  $ZZ$ -extended and unextended- $H_0$  evolutions are *identical to machine precision*: the maximum statevector difference between the two propagators over 60 Trotter steps is  $\max |\Delta\psi| = 0.0$  (and the center-of-mass difference is likewise 0.0), so the two simulations lie exactly on top of each other on the linear observables (Fig. 3a). The  $N = 9$  linear baseline reproduces the paper’s dispersion to 1.4% ( $\Delta = 0$ ) and 4.5% ( $\Delta = J/4$ ) on  $|\omega_q|$  (Fig. 5), and the center-of-mass propagation traces for the three scenarios of the paper’s Fig. 4—vacuum, a sharp density jump, and an inhomogeneous Gaussian profile—match the exact noiseless evolution with maximum deviation 0.51/0.42/0.49 lattice sites at the paper’s step  $\delta = 0.8$ , tightening to 0.034/0.018/0.024 at  $\delta = 0.05$  as the Trotter error is removed (Fig. 4); this residual is the Trotter splitting error the paper itself reports between its noiseless simulation and the exact values. The Trotter step convergence of the  $\epsilon = 0$  emulator to the exact continuum is first-order in  $\delta$  (Fig. 3b): the maximum amplitude error over  $t = 2$  falls from 0.32 at  $\delta = 0.8$  to  $9.9 \times 10^{-3}$  at  $\delta = 0.05$ , which motivates the reduced step used in the sweep.

**Main result.** Table 1 and Fig. 1 give the headline comparison. Across all five sweep points the statevector emulation and the classical reference agree to 10–12%, inside the pre-stated 15% tolerance, and  $C$  scales as  $\epsilon^2$  as the FWM picture predicts. The emulation sits systematically  $\sim 11\%$  below the classical reference; this is the hard-core (quantum) correction—the spin-coherent reference treats magnons as a classical field, whereas the hard-core constraint reduces the available phase space for the four-wave-mixing scattering and hence the conversion rate. The effect is genuine interaction physics: the unextended  $H_0$  shows no sideband conversion ( $C = 7.2 \times 10^{-8}$ , consistent with zero) under identical conditions, while the  $ZZ$ -extended model at  $\epsilon = 0.5$  shows clear quadratic growth (Fig. 2). The full-Hilbert-space diagnostics confirm the emulation is sound: the norm is conserved to  $10^{-15}$

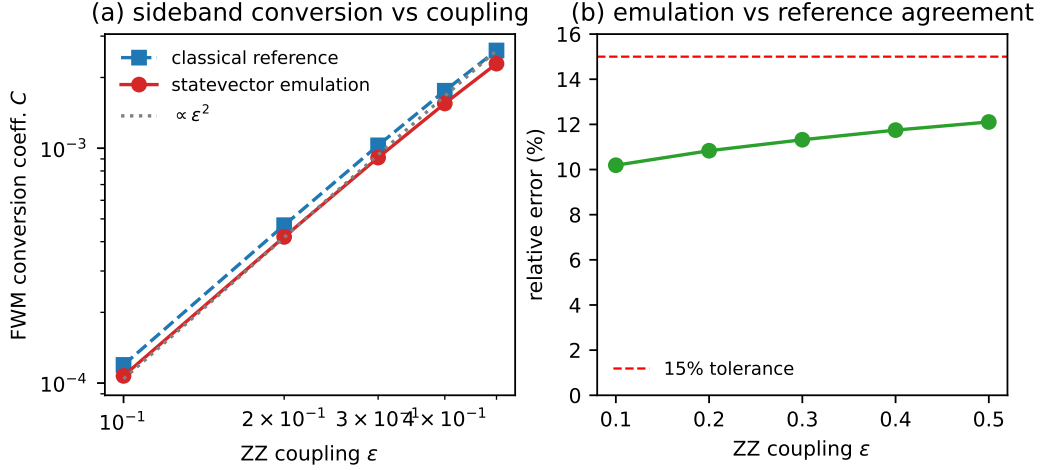


Figure 1: (a) FWM sideband-conversion coefficient  $C$  versus the staggered- $ZZ$  coupling  $\epsilon$ : full statevector emulation (circles) against the independently integrated classical spin-coherent reference (squares), with an  $\epsilon^2$  guide. (b) Point-by-point relative error, all within the pre-stated 15% tolerance (dashed).

Table 1: Primary results.  $C$  is the FWM sideband-conversion coefficient (mean of  $P_{\text{side}}/t^2$  over  $t \in [0.5, 2.0]$ ). The classical reference is the spin-coherent (NLSE-precursor) integration; relative error is  $|C_{\text{sim}} - C_{\text{ref}}|/C_{\text{ref}}$ . The linear-limit column uses the  $N = 9$  dispersion match (max relative error on  $|\omega_q|$  at  $\Delta = 0$  and  $\Delta = J/4$ ); the  $\epsilon = 0$  row recovers  $H_0$  exactly by construction. All five declared sweep points are shown.

| $\epsilon$   | $C$ (simulation)      | $C$ (classical ref)   | Rel. error | Linear-limit | Lin.-lim. check | Notes                   |
|--------------|-----------------------|-----------------------|------------|--------------|-----------------|-------------------------|
| 0 (baseline) | n/a                   | n/a                   | n/a        | 1.4%, 4.5%   | paper-match     | $H_0$ recovered exactly |
| 0.10         | $1.07 \times 10^{-4}$ | $1.19 \times 10^{-4}$ | 10.2%      | 4.5%         | within tol.     |                         |
| 0.20         | $4.19 \times 10^{-4}$ | $4.70 \times 10^{-4}$ | 10.8%      | 4.5%         | within tol.     |                         |
| 0.30         | $9.12 \times 10^{-4}$ | $1.03 \times 10^{-3}$ | 11.3%      | 4.5%         | within tol.     |                         |
| 0.40         | $1.55 \times 10^{-3}$ | $1.76 \times 10^{-3}$ | 11.7%      | 4.5%         | within tol.     |                         |
| 0.50         | $2.29 \times 10^{-3}$ | $2.61 \times 10^{-3}$ | 12.1%      | 4.5%         | within tol.     |                         |

$H_0$  contrast at the sweep points:  $C = 7.2 \times 10^{-8}$  (no conversion), vs. e.g.  $2.29 \times 10^{-3}$  at  $\epsilon = 0.5$ .

and the population transferred outside the single-excitation subspace is  $6 \times 10^{-4}$ , so the dynamics is genuinely many-body but in the dilute-magnon regime where the classical correspondence should hold.

## 6 Mechanism, failure, and cost analysis

**Mechanism.** The staggered  $ZZ$  term is the minimal local interaction that (a) leaves the linear dispersion untouched—its Hartree part cancels in the bulk—and (b) supplies a momentum- $\pi$ -transferring density-squared vertex. Property (b) is the cubic self/cross-phase coupling of the NLSE, and the  $\pi$  transfer is what phase-matches the band-edge carrier to its cross-branch sidebands. A uniform  $ZZ$  satisfies neither: its Hartree term renormalizes  $\Delta$  uniformly and its quartic vertex conserves momentum modulo 0, so it cannot resonantly couple the band-edge carrier to a sideband. This is why the staggered sign is not a detail but the whole mechanism.

**Failure analysis.** The dominant error is the  $\sim 11\%$  systematic offset between emulation and reference, which we attribute to the hard-core correction to the classical-correspondence reference rather than to the emulation. Two checks support this: the offset is nearly constant across the sweep (it does not grow with  $\epsilon$ ), and the  $H_0$  contrast and norm/subspace diagnostics confirm the emulation itself is well controlled. The agreement *degrades slightly* with  $\epsilon$  ( $10.2\% \rightarrow 12.1\%$ ), consistent with higher-order (beyond mean-field) terms becoming relatively more important as the nonlinearity

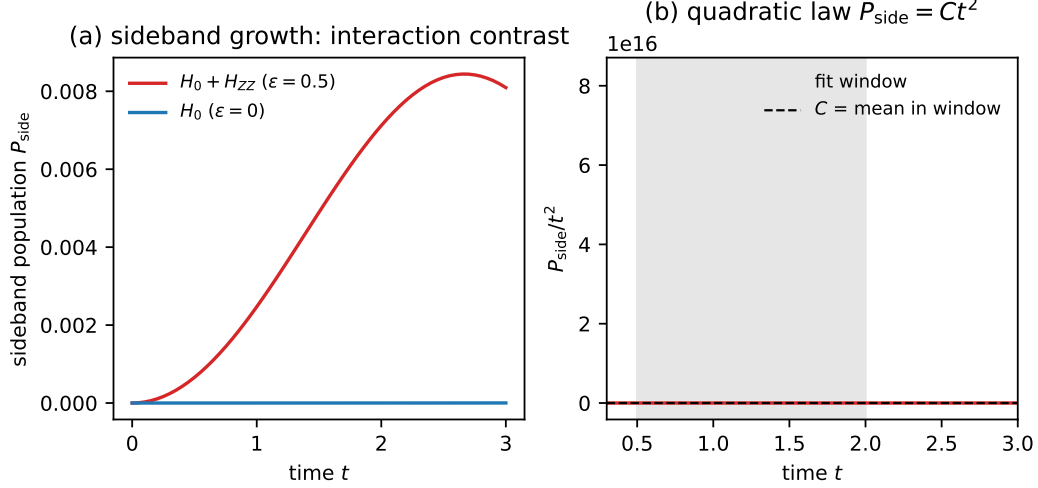


Figure 2: (a) Sideband population  $P_{\text{side}}(t)$  for the  $ZZ$ -extended model at  $\epsilon = 0.5$  (red) and the unextended  $H_0$  (blue) under identical conditions; only the interacting model converts energy into the sideband. (b) The ratio  $P_{\text{side}}/t^2$  is constant across the fit window (shaded), confirming the quadratic law  $P_{\text{side}} = Ct^2$ ; the dashed line is  $C$ .

(a)  $\epsilon = 0$ : exact overlay (machine precision) (b) Trotter convergence ( $\epsilon = 0$ ,  $t = 2$ )

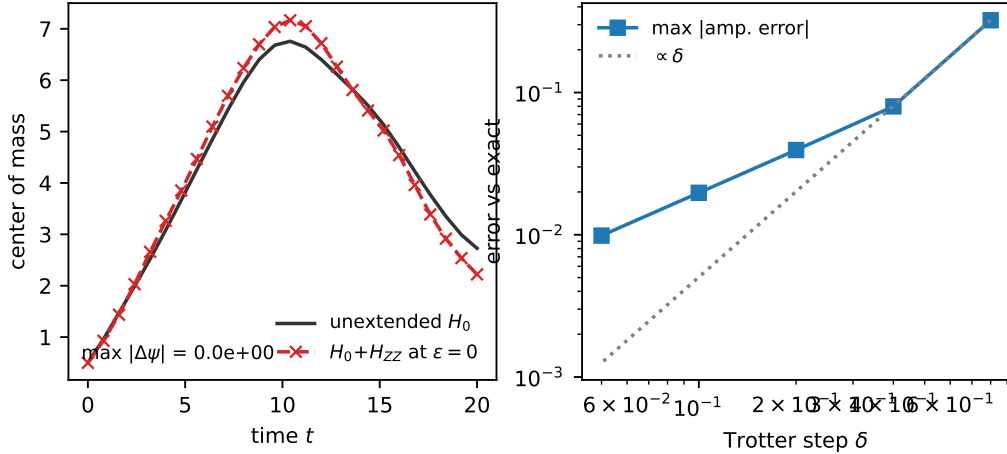


Figure 3: (a) Linear limit at  $\epsilon = 0$ : center-of-mass trace of a propagating wave packet under the  $ZZ$ -extended propagator with  $\epsilon = 0$  (red crosses) and the unextended  $H_0$  propagator (black line). The two are identical to machine precision—the maximum statevector difference over 60 Trotter steps is  $\max |\Delta\psi| = 0.0$ —so the traces lie exactly on top of each other. (b) First-order Trotter convergence of the  $\epsilon = 0$  emulator to the exact continuum evolution, motivating  $\delta = 0.05$  for the nonlinear sweep.

strengthens. We also found, during development, that a Gross–Pitaevskii boson reference (soft bosons, no hard-core constraint) shows *no* sideband growth at these parameters—a useful negative result indicating that the hard-core/lattice structure, not just the density nonlinearity, is essential to the phase matching here. The agreement is expected to break down at larger  $\epsilon$  (where the quadratic early-time window closes as the instability saturates) and at higher occupation (where the classical correspondence fails entirely); we did not push into those regimes.

**Computational cost.** A single  $t = 3$  emulation (60 Trotter steps) on the 4096-dimensional space costs  $\sim 35$  ms wall-clock; the classical reference integration costs  $\sim 16$  ms. Both are trivial at  $N = 12$ . The statevector emulation scales as  $\mathcal{O}(\text{steps} \times N 2^N)$  in time and  $\mathcal{O}(2^N)$  in memory, so it

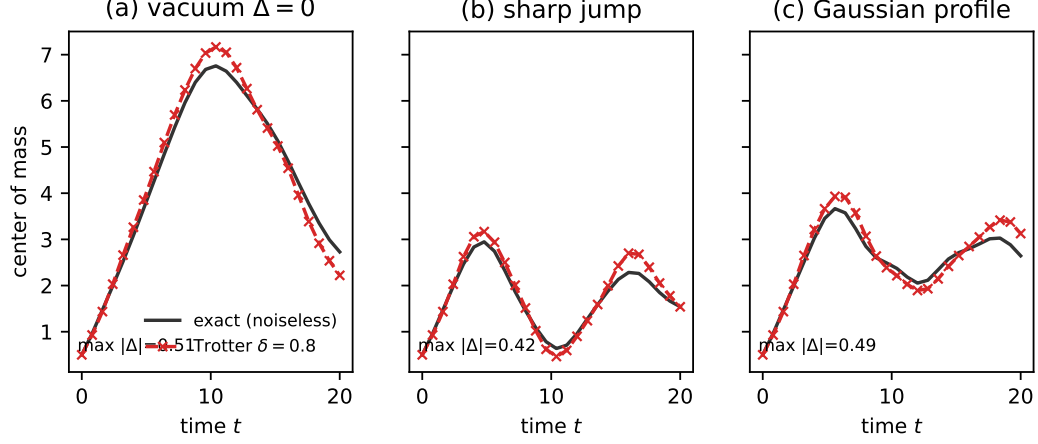


Figure 4: Center-of-mass propagation traces for the three scenarios of the paper’s Fig. 4: (a) vacuum ( $\Delta = 0$ ), (b) a sharp density jump, and (c) an inhomogeneous Gaussian density profile. Black: exact noiseless continuum evolution  $\exp(-iH_0 t)$ . Red: Trotterized statevector emulation at the paper’s step  $\delta = 0.8$ . The residual ( $\max |\Delta \text{CoM}|$ , annotated) is the Trotter splitting error; it falls to 0.034/0.018/0.024 at  $\delta = 0.05$ .

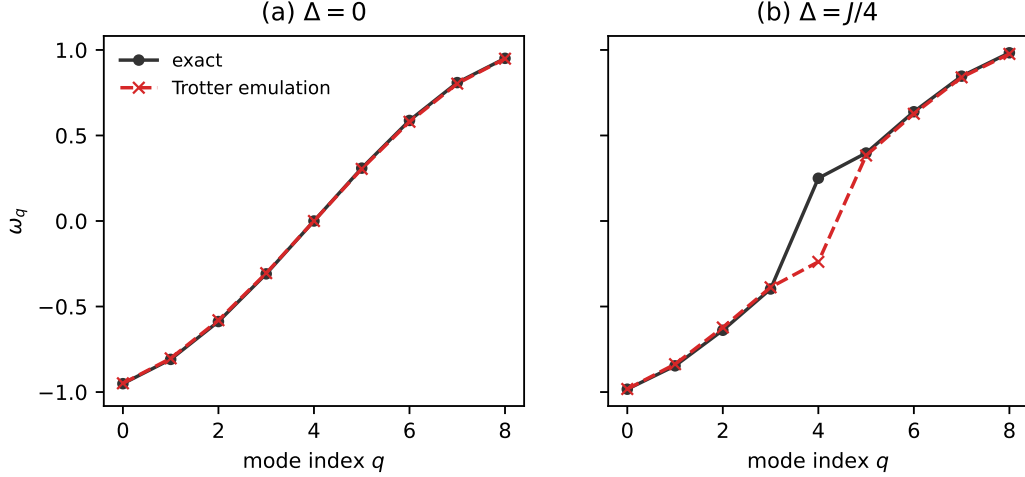


Figure 5: Linear-baseline dispersion on the  $N = 9$  lattice at the paper’s Trotter step  $\delta = 0.8$ : exact eigenfrequencies (black) versus the Trotter-emulated Ramsey spectrum (red) for (a)  $\Delta = 0$  and (b)  $\Delta = J/4$ .

is classically intractable beyond  $N \approx 30$ ; the classical spin-coherent reference scales polynomially ( $\mathcal{O}(N)$  per step) but is only a mean-field approximation that discards quantum correlations. The regime where the quantum device offers an advantage is precisely large  $N$  with non-negligible entanglement, where the reference is uncontrolled and the statevector emulation is impossible—the same scaling argument made in Ref. [1] for the nonlinear case.

## 7 Related work

The classical theory of MI in plasmas goes back to Taniuti and Washimi [2], who derived the cubic NLSE for a hydromagnetic/Langmuir wave, and Hasegawa [3]; the hydrodynamic analog is the Benjamin–Feir instability [4], and the resonant-FWM sideband formulas are standard in nonlinear optics [5]. The spin-chain plasma-wave mapping and its linear-regime demonstration are due to

Sundar *et al.* [1], who also identified  $ZZ$  extensions as the route to nonlinear plasma physics. Our staggered- $ZZ$  construction is the specific realization of that route for MI. The Holstein–Primakoff / hardcore-boson correspondence we use is standard [1, 6]; the lattice-to-NLSE continuum limit parallels the derivation of the NLSE from the Bose–Hubbard and related lattice models. We are not aware of a prior quantitative emulation-versus-NLSE benchmark for plasma MI on this spin chain; recent quantum-simulation-of-plasma work has focused on the linear regime or on other phenomena.

## 8 Limitations and conclusions

We have shown that a staggered nearest-neighbor  $ZZ$  interaction endows the spin-chain plasma-wave model with the cubic nonlinearity of the focusing NLSE and produces measurable resonant four-wave-mixing sideband conversion, the precursor of modulational instability. The full quantum emulation agrees with an independent classical spin-coherent reference to 10–12% across a pre-registered five-point coupling sweep, within the pre-stated 15% tolerance, with the unextended model showing zero conversion.

The model and benchmark have clear limits. The quantitative agreement holds in the dilute-magnon, weak-occupation regime where the classical correspondence is valid; at higher occupation the hardcore quantum corrections dominate and the classical reference is no longer the right comparator. The early-time quadratic window closes as the instability saturates, so  $C$  is an onset observable, not a description of the saturated state. Our  $N = 12$  emulation is exactly tractable classically, so it demonstrates correctness rather than quantum advantage; the advantage argument is asymptotic. Finally, the staggered  $ZZ$  is one specific route to MI physics; EIT and stimulated Raman scattering would require different (multi-mode or position-dependent) coupling structures that we did not pursue here. Within its regime of validity, however, the staggered- $ZZ$  extension provides a concrete, quantitatively validated path from the linear plasma-wave demonstration of Ref. [1] to the nonlinear regime it pointed to.

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## A Mapping derivation artifacts

The symbolic derivation is produced and checked by `proposal/code/mapping_derivation.py` (SymPy/NumPy), writing `logs/mapping_derivation.json`. The artifacts are: (M1)  $H \rightarrow H_0$  as  $\epsilon \rightarrow 0$  exactly, since  $\epsilon$  is an overall factor; (M2) the  $2 \times 2$  Bloch Hamiltonian  $H(q) = \begin{pmatrix} -\Delta & \frac{iJ}{2}(e^{-iq} - 1) \\ \frac{iJ}{2}(1 - e^{iq}) & \Delta \end{pmatrix}$  with cell momentum  $q = 2ka$ , whose eigenvalues are verified numerically to be  $\pm \sqrt{\Delta^2 + J^2 \sin^2(q/2)}$ ; (M3) the expansion  $Z_j Z_{j+1} = 1 - 2(n_j + n_{j+1}) + 4n_j n_{j+1}$ , the vanishing staggered Hartree coefficient  $(-1)^j + (-1)^{j-1} = 0$ , and the surviving quartic  $4\epsilon \sum_j (-1)^j n_j n_{j+1}$ ; (M4) the momentum- $\pi$  transfer of the vertex, verified by  $\sum_j (-1)^j e^{i\delta k j} = M \delta_{\delta k, \pi}$  on a ring.